



US012406267B2

(12) **United States Patent**
Doebler et al.

(10) **Patent No.:** **US 12,406,267 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SUPPLIER DATA VALIDATION USING
VALIDATION SUBSYSTEM**

(71) Applicant: **Coupa Software Incorporated**, San
Mateo, CA (US)
(72) Inventors: **Kathryn C. Doebler**, Sewickley, PA
(US); **Oleksandr Mekhovov**, Hamburg
(DE); **Faraz A. Qureshi**, Astoria, NY
(US); **Ahmad Sadeddin**, Daly City, CA
(US); **Kengsreng Tang**, Hayward, CA
(US); **Himica Kumar**, San Francisco,
CA (US)

(73) Assignee: **Coupa Software Incorporated**, Foster
City, CA (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **17/186,900**

(22) Filed: **Feb. 26, 2021**

(65) **Prior Publication Data**

US 2022/0277314 A1 Sep. 1, 2022

(51) **Int. Cl.**
G06Q 30/018 (2023.01)
G06F 16/23 (2019.01)
G06Q 20/10 (2012.01)
G06F 3/0481 (2022.01)

(52) **U.S. Cl.**
CPC **G06Q 30/018** (2013.01); **G06F 16/2365**
(2019.01); **G06Q 20/10** (2013.01); **G06F**
3/0481 (2013.01)

(58) **Field of Classification Search**
CPC G06Q 30/018; G06Q 30/00
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,689,482 B2	3/2010	Lam et al.	
7,742,985 B1	6/2010	Digrigoli et al.	
7,970,671 B2	6/2011	Hahn-Carlson et al.	
8,275,705 B2 *	9/2012	Katz	G06Q 20/389 705/40
RE44,502 E	9/2013	Boesch et al.	

(Continued)

FOREIGN PATENT DOCUMENTS

WO 2008/042823 4/2008

Primary Examiner — Calvin L Hewitt, II

Assistant Examiner — Kevin T Poe

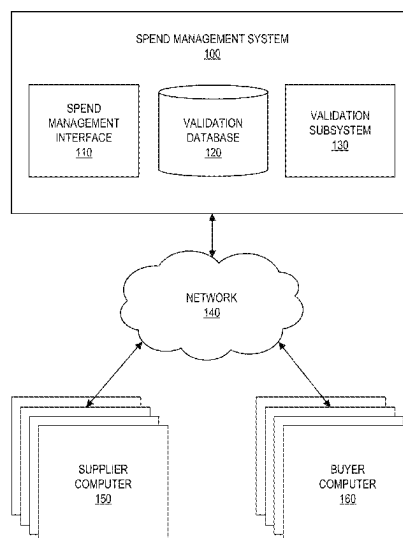
(74) *Attorney, Agent, or Firm* — Baker Botts L.L.P.

(57) **ABSTRACT**

A computer-implemented method for validating supplier data is provided. The method includes accessing a spend management interface of a spend management system. The spend management interface may receive supplier data from a supplier computer via user interface widgets of the spend management interface. A country identifier is received from the supplier computer via a country widget and a currency identifier is received via a currency widget of the user interface widgets. One or more parameters associated with the country identifier and currency identifier are identified within a validation database of the spend management system and customized user interface widgets are generated based on the one or more parameters. The spend management interface is electronically updated to present the customized user interface widgets to the supplier computer. Supplier data is received from the supplier computer via the customized user interface widgets and the supplier data is programmatically validated.

9 Claims, 6 Drawing Sheets

170



(56)

References Cited

U.S. PATENT DOCUMENTS

10,489,753	B2	11/2019	Allen-Rouman et al.	
10,810,582	B2	10/2020	Digrigoli et al.	
11,311,797	B2 *	4/2022	Rose	G06Q 20/385
2002/0153415	A1 *	10/2002	Minami	G06Q 20/18 235/380
2013/0151417	A1 *	6/2013	Gupta	G06Q 20/382 705/65
2014/0222594	A1 *	8/2014	Rose	G06Q 20/385 705/16
2015/0120512	A1 *	4/2015	Chen	G06Q 30/0641 705/27.1
2016/0054908	A1 *	2/2016	Vegesna	G06F 3/04842 715/716
2016/0071069	A1 *	3/2016	Skala	G06Q 20/4016 705/32
2016/0357526	A1 *	12/2016	Soffer	G06F 8/36
2020/0094133	A1 *	3/2020	Rose	G06Q 20/385
2020/0259903	A1 *	8/2020	Momchilov	H04L 67/025
2021/0397681	A1 *	12/2021	Boule	G06F 21/83

* cited by examiner

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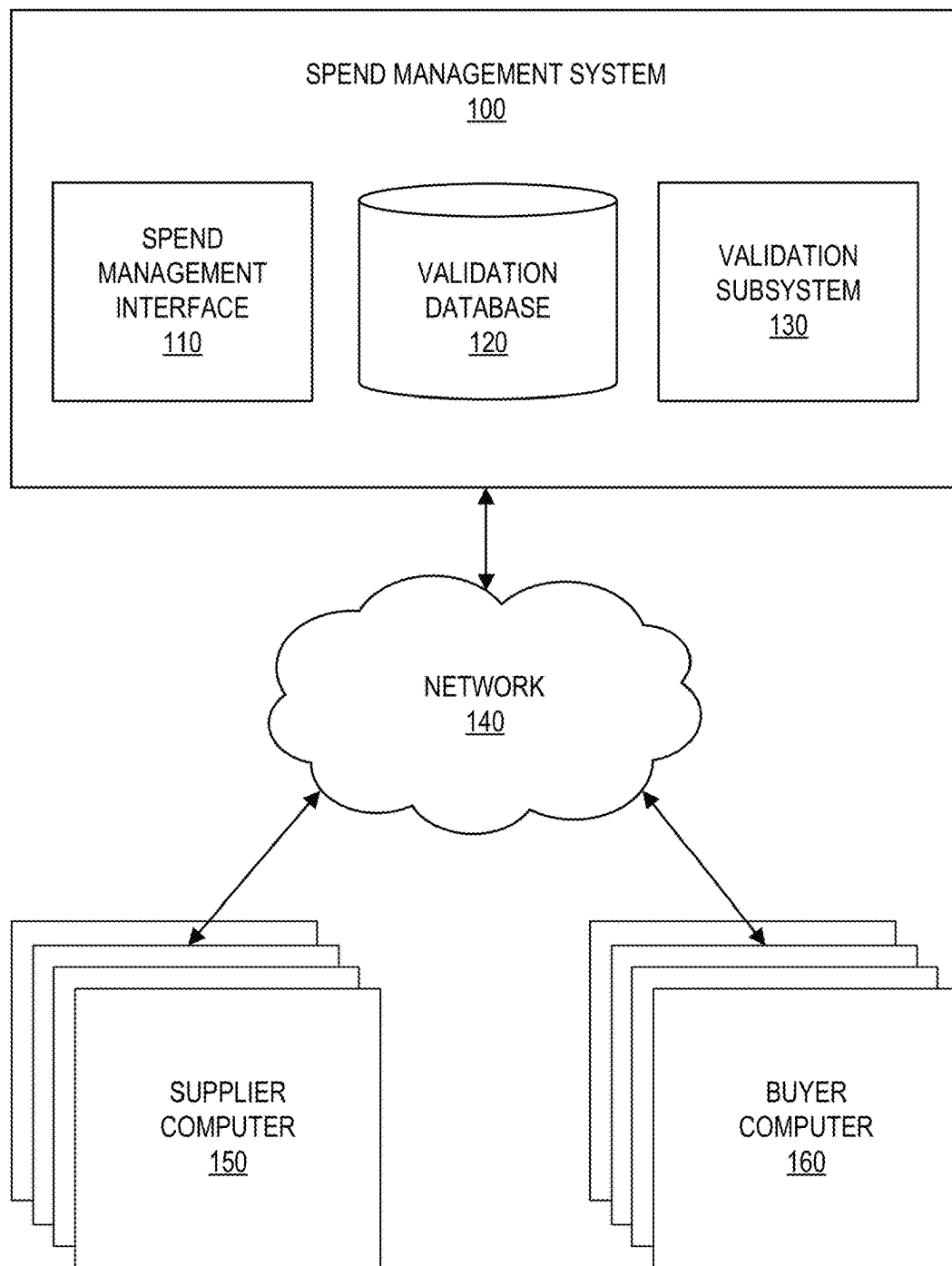


FIG. 1

110

Payment Method

* Payment Type: Bank Account

What are your Bank Account Details?

200 → Bank Account Country/Region: United States ← 201

202 → Bank Account Currency: [Dropdown]

204 { Beneficiary Name: [Text Field]
Bank Name: [Text Field]
Account Number: [Text Field]
Confirm Account Number: [Text Field]
Routing Number: [Text Field]
SWIFT/BIC Code: [Text Field]
Bank Account Type: [Text Field]

What is your Bank's Branch Address?

Address Line 1: [Text Field]
Address Line 2: [Text Field]
City: [Text Field]
State: Select an Option [Dropdown]
Postal Code: [Text Field]

FIG. 2A

110

Payment Method

Payment Type Bank Account

What are your Bank Account Details?

200 Bank Account Country/Region United States 201

202 Bank Account Currency USD 203

204 Beneficiary Name Armand Co.

Bank Name

Account Number 208

Cardless Account Number 210

206 Routing Number

SWIFT/BIC Code

Bank Account Type Business

Please enter the Account Number (16 to 17 digits)

What is your Bank's Branch Address?

Address Line 1

Address Line 2

City

State United States

Postal Code

FIG. 2B

110

Payment Method

Payment Type Bank Account

What are your Bank Account Details?

200 Bank Account Country/Region United States 201

202 Bank Account Currency EUR 203

204 Beneficiary Name Armand Co.

Bank Name

Account Number

Cardless Account Number

IBAN

Cardless IBAN

214 Routing (Bank Code) Number Routing no

SWIFT/BIC Code

Bank Account Type Business

What is your Bank's Branch Address?

Address Line 1

Address Line 2

City

State United States

FIG. 2C

110

Payment Method

* Payment Type: Bank Account

What are your Bank Account Details?

200 → Bank Account Country/Region: United States ✓ ← 201

202 → Bank Account Currency: USD ✓ ← 203

204 { Beneficiary Name: Account On

Bank Name:

Account Number: *

Confirm Account Number:

300 → Routing Number: 1234567890 *

302 → must be 9 digits

SWIFT/BIC Code: *

Bank Account Type: Business

What is your Bank's Branch Address?

Address Line 1:

Address Line 2:

City:

State: California

Postal Code:

FIG. 3A

110

* Payment Type: Bank Account

What are your Bank Account Details?

200 → Bank Account Country/Region: Australia ✓ ← 201

202 → Bank Account Currency: AUD ✓ ← 203

204 { Beneficiary Name:

Bank Name:

Account Number: *

Confirm Account Number:

304 → 999: 123 *

306 → must be 6 digits

SWIFT/BIC Code:

Bank Account Type: Business

208

210

FIG. 3B

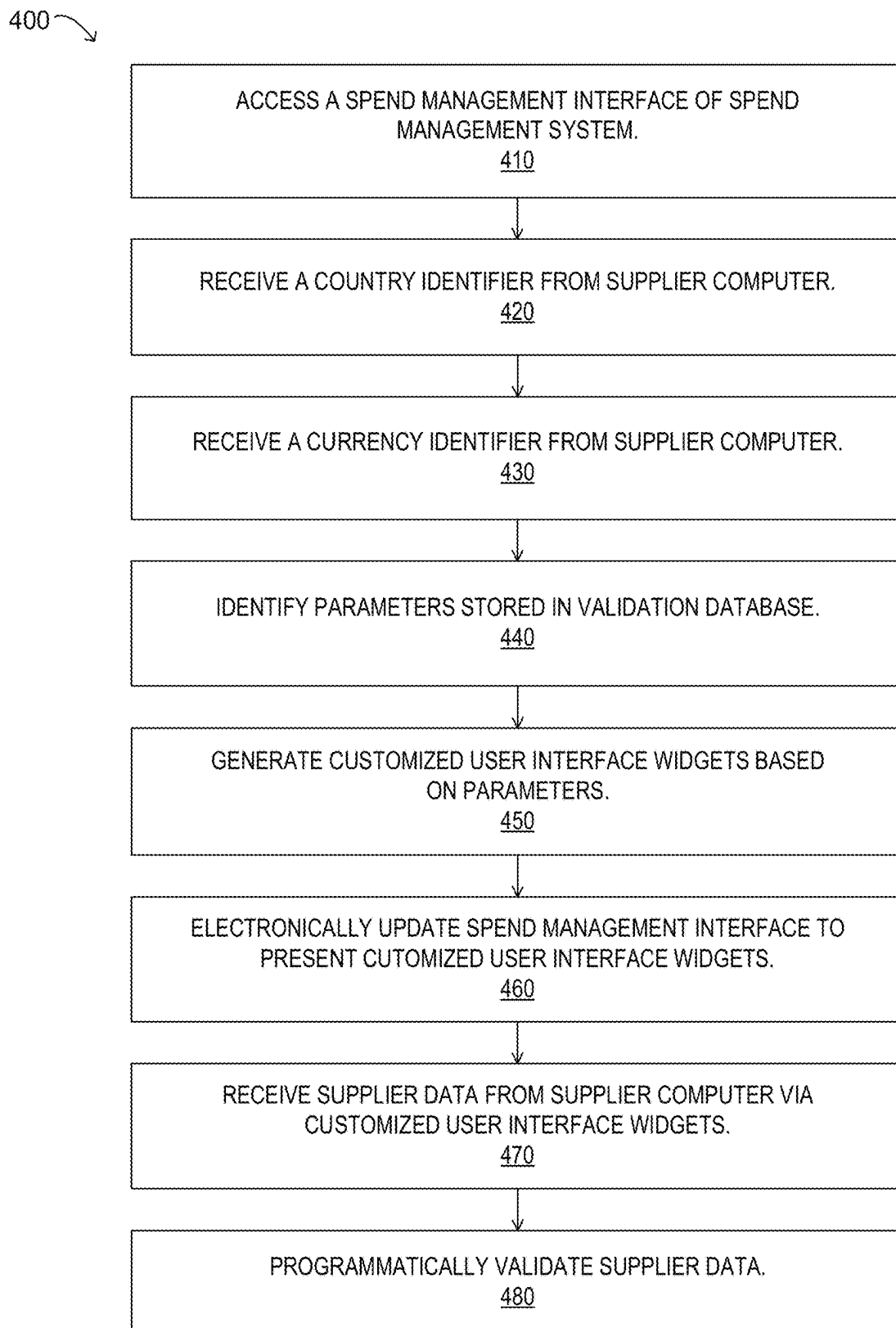


FIG. 4

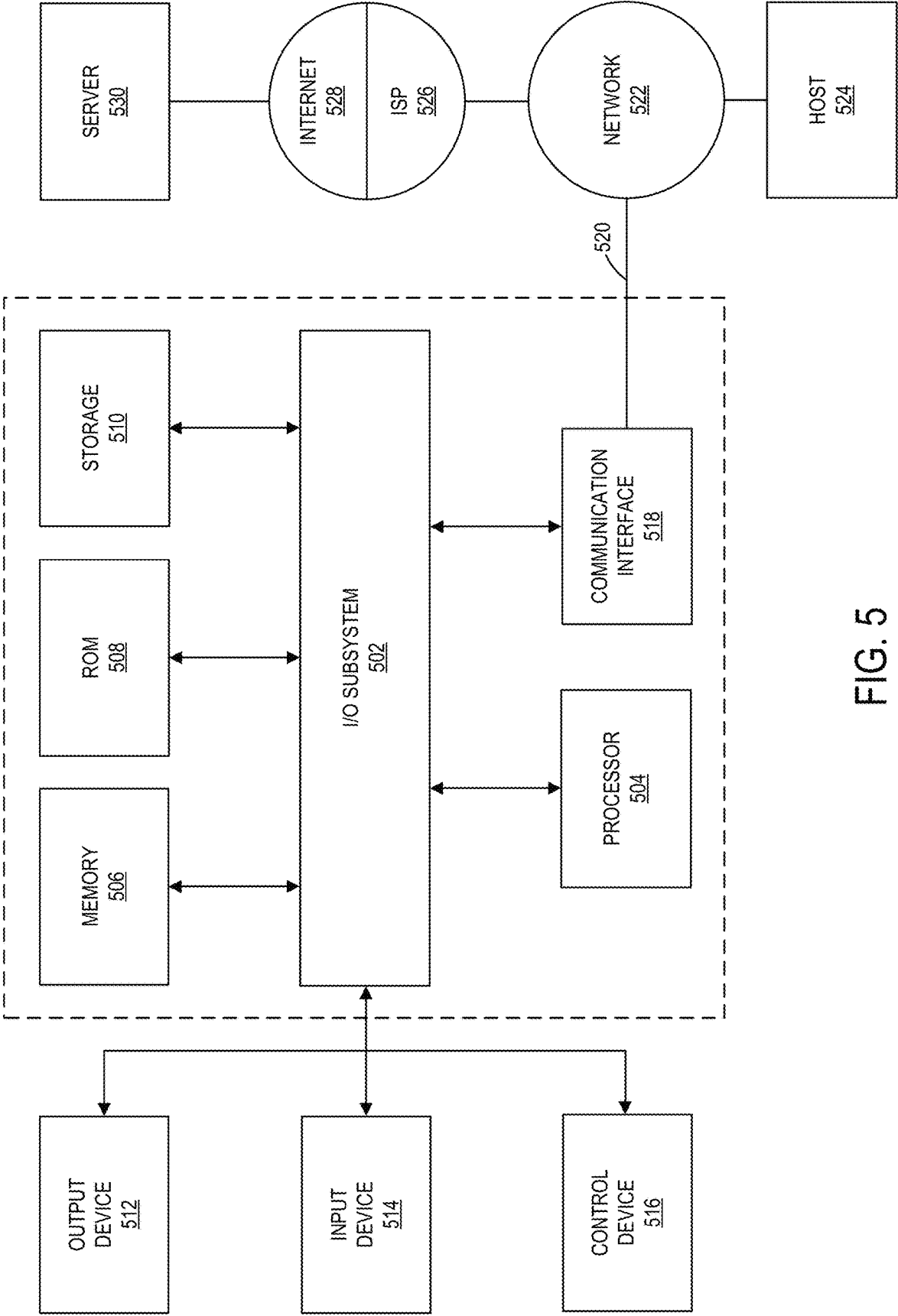


FIG. 5

SUPPLIER DATA VALIDATION USING VALIDATION SUBSYSTEM

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FIELD OF THE DISCLOSURE

One technical field of the present disclosure is computer-implemented supplier data validation using a validation subsystem. Another field is computer-implemented supplier data management using a validation subsystem. Yet another technical field is computer-implemented electronic presentation of customized user interface widgets via a spend management interface.

BACKGROUND

The approaches described in this section are approaches that could be pursued, but not necessarily approaches that have been previously conceived or pursued. Therefore, unless otherwise indicated, it should not be assumed that any of the approaches described in this section qualify as prior art merely by virtue of their inclusion in this section. Further, it should not be assumed that any of the approaches described in this section are well-understood, routine, or conventional merely by virtue of their inclusion in this section.

Computer-implemented software-as-a-service (SaaS) systems are now available for a variety of data processing applications. These systems typically offer complex, sophisticated internal data processing applications. For example, e-procurement or spend management systems have been developed to assist enterprises in tracking and planning purchasing and sourcing operations at high volume. Suppliers may use these systems to provide supplier bank account data, or “supplier data,” to buyers for payment for goods or services. Specifically, suppliers may create respective digital repositories within spend management systems that digitally store the supplier data necessary for buyers to complete bank transactions. For example, such digital repositories may store supplier bank account details such as bank codes, routing numbers, account numbers, and/or other supplier data that buyers may use to address supplier bank accounts for bank account transactions.

Typically, suppliers may manually provide supplier data to spend management systems. For example, suppliers may manually enter supplier data into a spend management system by accessing one or more data fields electronically presented within a user interface of the spend management system using a supplier computer. However, such manual entry of supplier data—which may arrive at the spend management system in large quantities, such as thousands or millions of data entries—may be validated by spend management systems using a generic set of syntax requirements, often resulting in missing, incomplete, and/or incorrect supplier data being stored in the spend management system. Further, spend management systems may use the generic set

of syntax requirements to validate supplier data uniformly across various countries and currency types.

However, using a generic set of syntax requirements to validate supplier data neglects to account for specific syntax requirements corresponding to a given country in which a supplier bank account is located and/or a given currency type associated with a supplier bank account transaction. For example, a supplier bank account located in the United States may require a routing number to complete a supplier bank account transaction using United States dollars (USD) while a supplier bank account located in Germany may require an International Bank Account Number (IBAN) to complete a supplier bank account transaction using euros (EUR). In this example, the spend management system may electronically present one or more data fields within a user interface accessed by a supplier computer that may, or may not, represent the specific syntax requirements corresponding to the country in which the supplier bank account is located and/or the given currency type associated with the supplier bank account transaction. Here, suppliers may manually enter inaccurate supplier data into the one or more data fields, which may result in failures to accurately integrate supplier data across multiple spend management systems accessed by buyers (e.g., enterprise resource planning (ERP) platforms) for making payments to suppliers. Inaccurate supplier data may additionally result in costly bounced payments from banks due to buyers attempting to use the inaccurate supplier data for supplier bank account transactions. Or, inaccurate supplier data may result in suppliers not receiving payments for products or services provided to buyers.

In some instances, inaccurate supplier data may be detected and mitigated by the buyer before payments are made. However, in an environment involving thousands to millions of suppliers located across multiple countries and requiring various currency types, merely detecting the inaccurate supplier data becomes a time-consuming, complex problem. Wide variances in bank account transaction requirements for respective countries and currency types mean that buyers may be motivated to introduce new mechanisms for supplier data validation (e.g., third-party software), thereby exposing sensitive data which presents opportunities for fraud and/or additional disruptions in supplier bank account transactions. Furthermore, the use of inaccurate supplier data for bank account transactions invariably causes the systems involved in implementing an e-procurement system, such as server instances and client computers, to use excess amounts of computing resources such as CPU cycles, storage, memory, and/or network bandwidth that may be avoidable. Thus, there is a need for ways to electronically present one or more data fields within a user interface of a spend management system accessed by supplier computers that each represent the specific syntax requirements corresponding to the country in which a supplier bank account is located and/or the given currency type associated with a supplier bank account transaction. Such specific syntax requirements may be used to validate supplier data entered manually into the one or more data fields by suppliers, thereby substantially reducing the time and complexity involved in validating supplier data stored in database records of a spend management system.

SUMMARY OF THE INVENTION

The appended claims may serve as a summary of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

In the drawings:

FIG. 1 is a block diagram of an example networked computer system in which various embodiments may be practiced.

FIG. 2A, FIG. 2B, FIG. 2C illustrate an example spend management interface of a spend management system, according to various embodiments.

FIG. 3A, FIG. 3B illustrate an example spend management interface that includes an error notification, according to various embodiments.

FIG. 4 illustrates a flow diagram of an algorithm that may be programmed to implement an embodiment.

FIG. 5 illustrates a computer system that may be used to implement embodiments.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be apparent, however, that the present invention may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring the present invention. Embodiments are described herein in sections according to the following outline:

1. General Overview
2. Structural Overview
 - 2.1 Example Distributed Computer System
 - 2.2 User Interface Example
 - 2.3 Supplier Data Validation Example
3. Functional Overview
4. Implementation Example—Hardware Overview
5. Benefits; Extensions and Alternatives

1. General Overview

Techniques for validating and managing supplier data using a validation subsystem are described herein. According to an embodiment, the validation subsystem may access a spend management interface of a spend management system. The spend management interface may be programmed to receive supplier data from a supplier computer via one or more user interface widgets of the spend management interface. The validation subsystem may receive a country identifier from the supplier computer via a country widget of the one or more user interface widgets. The country identifier may indicate a country in which a supplier bank account is located. The validation subsystem may receive a currency identifier from the supplier computer via a currency widget of the one or more user interface widgets. The currency identifier may indicate a currency type associated with a supplier bank account transaction. The validation subsystem may identify one or more parameters of bank account data parameters stored in a validation database of the spend management system. The one or more parameters may be associated with the country identifier and the currency identifier. The validation subsystem may generate one or more customized user interface widgets based on the

one or more parameters. The one or more customized user interface widgets may be associated with all data values of the supplier data necessary to correctly address the supplier bank account for the supplier bank account transaction. The validation subsystem may electronically update the spend management interface to present the one or more customized user interface widgets to the supplier computer and may receive the supplier data from the supplier computer via the one or more customized user interface widgets. The validation subsystem may then programmatically validate the supplier data.

In an embodiment, a method comprises, using a validation subsystem of a computer-implemented spend management system, accessing a spend management interface of the spend management system, the spend management interface being programmed to receive supplier data from a supplier computer via one or more user interface widgets of the spend management interface; receiving a country identifier from the supplier computer via a country widget of the one or more user interface widgets, the country identifier indicating a country in which a supplier bank account is located; receiving a currency identifier from the supplier computer via a currency widget of the one or more user interface widgets, the currency identifier indicating a currency type associated with a supplier bank account transaction; identifying one or more parameters of a plurality of bank account data parameters stored in a validation database of the spend management system, the one or more parameters associated with the country identifier and the currency identifier; generating one or more customized user interface widgets based on the one or more parameters, the one or more customized user interface widgets being associated with all data values of the supplier data necessary to correctly address the supplier bank account for the supplier bank account transaction; electronically updating the spend management interface to present the one or more customized user interface widgets to the supplier computer; receiving the supplier data from the supplier computer via the one or more customized user interface widgets; and programmatically validating the supplier data.

2. Structural Overview

2.1 Example Distributed Computer System

Computer executable instructions described herein may be in machine executable code in the instruction set of a CPU and may have been compiled based upon source code written in JAVA, C, C++, OBJECTIVE-C, or any other human-readable programming language or environment, alone or in combination with scripts in JAVASCRIPT, other scripting languages and other programming source text. In another embodiment, the programmed instructions also may represent one or more files or projects of source code that are digitally stored in a mass storage device such as non-volatile RAM or disk storage, in the systems of FIG. 1 or a separate repository system, which when compiled or interpreted cause generating executable instructions which when executed cause the computer to perform the functions or operations that are described herein with reference to those instructions. In other words, the drawing figure may represent the manner in which programmers or software developers organize and arrange source code for later compilation into an executable, or interpretation into bytecode or the equivalent, for execution by the spend management system 100.

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FIG. 1 is a block diagram of an example networked computer system in which various embodiments may be practiced. In the embodiment illustrated in FIG. 1, a networked computer system 170 comprises a spend management system 100, supplier computers 150 (individually referred to herein as “supplier computer 150”), and buyer computers 160 (individually referred to herein as “buyer computer 160”), which are communicatively coupled directly or indirectly via network 140. The spend management system 100 may include a spend management interface 110, a validation database 120, and a validation subsystem 130 which are communicatively coupled via one or more direct or indirect digital electronic network links, using any of wired or wireless, terrestrial or satellite network links, internetworks, and infrastructure gear. The spend management system 100, supplier computers 150, buyer computers 160, and other elements of the networked computer system 170 may each comprise an interface compatible with the network 140 and are programmed or configured to use standardized protocols for communication across the networks such as TCP/IP, Bluetooth, and higher-layer protocols such as HTTP, TLS, and the like. FIG. 1 is shown in simplified, schematic format for purposes of illustrating a clear example and other embodiments may include additional, fewer, and/or different elements.

The computing devices such as the supplier computers 150 and buyer computers 160 may comprise a desktop computer, laptop computer, tablet computer, smartphone, or any other type of computing device that allows access to the spend management system 100. In one embodiment, suppliers may access the spend management system 100 using the supplier computers 150 to provide supplier data for receiving payments from buyers associated with buyer computers 160 (e.g., companies, private entities, business enterprises, and the like). Specifically, the supplier computers 150 may access the spend management system 100 (e.g., via spend management interface 110) such that suppliers may manually provide the supplier data necessary for buyers to address supplier bank accounts for completing bank account transactions. For example, suppliers may manually provide supplier data to the spend management system 100 such as bank codes, routing numbers, and/or account numbers. Buyers may use respective buyer computers 160 to access supplier data stored in the spend management system 100 and to use the supplier data for completing bank account transactions accordingly. In one embodiment, supplier computers 150 may provide inaccurate supplier data for buyers associated with buyer computers 160. That is, a supplier may provide inaccurate supplier data to buyers inadvertently by manually entering supplier data into the spend management system 100.

In one embodiment, the spend management system 100 may be programmed or configured for generating electronic pages, alerts, notifications, hyperlinks, recommendations, or application protocol messages to output to the computing devices such as the supplier computers 150 or buyer computers 160. For example, spend management system 100 may be programmed to generate dynamic web pages that implement an application as a software-as-a-service (SaaS) application that the supplier computer 150 and buyer computer 160 may access using a web browser hosted at the computing device. In one embodiment, the spend management system 100 may be programmed to generate a dynamic web page (e.g., spend management interface 110) used to electronically present supplier data to supplier computers 150 and/or buyer computers 160. Specifically, the spend management system 100 may be programmed to receive

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supplier data from supplier computers 150 and to programmatically validate the supplier data such that buyer computers 160 may access the validated supplier data to complete bank account transactions. In one embodiment, the spend management system 100 may be implemented using a server-class computer or other computers having one or more processor cores, co-processors, or other computers. Spend management system 100 may be a physical server computer and/or a virtual server instance stored in a data center, such as through cloud computing.

In one embodiment, the spend management interface 110 may comprise a system, device, or apparatus generally operable to electronically receive and present supplier data to supplier computers 150 and/or buyer computers 160. In particular, spend management interface 110 may be or include a dynamic web page that electronically presents one or more data fields, or “user interface widgets,” to supplier computers 150 such that suppliers may manually enter supplier data into the spend management system 100. In one embodiment, respective suppliers may manually provide, indicate, or otherwise specify, a country in which payments are to be received using the user interface widgets electronically presented within the spend management interface 110. Specifically, respective suppliers may indicate a country in which a supplier bank account is located by providing the name of the country, or a “country identifier,” into a data field configured to receive the country identifier, or a “country widget,” of the user interface widgets electronically presented within the spend management interface 110. In another embodiment, respective suppliers may manually provide, indicate, or otherwise specify, a currency type in which payments are to be received using the user interface widgets electronically presented within the spend management interface 110. In particular, respective suppliers may indicate a currency type associated with a supplier bank account transaction by providing the name of the currency, or a “currency identifier,” into a data field configured to receive the currency identifier, or a “currency widget,” of the user interface widgets electronically presented with the spend management interface 110. The spend management system 100 may be programmed to receive the supplier data provided by supplier computers 150 via the spend management interface 110 and to programmatically validate the supplier data such that buyers may use the supplier data for completing bank account transactions accordingly.

In one embodiment, the spend management interface 110 may additionally include one or more icons, or “hint icons,” disposed proximate to one or more user interface widgets. Each hint icon electronically presented within the spend management interface 110 may be programmed to provide suppliers with hints, or “suggestions,” regarding proper syntax, format, and/or length of supplier data to be manually entered by suppliers. Each suggestion provided to suppliers using hint icons may correspond to the user interface widget next to which the hint icon is presented. Hint icons are described in further detail with respect to FIG. 2B.

In one embodiment, the validation database 120 may comprise a system, device, or apparatus generally operable to digitally store one or more parameters, or “bank account data parameters,” associated with successfully completing bank account transactions in various countries using multiple currency types. Specifically, the validation database 120 may be managed by the spend management system 100 and may store the bank account data parameters within one or more entries, or “country block entries,” corresponding to respective country identifiers. In one embodiment, bank account data parameters of a given country block entry

stored in validation database **120** may correspond to a country in which a supplier bank account is located (e.g., as indicated by a country identifier). For example, bank account data parameters of a country block entry for the United States may include account number and routing number parameters, thereby indicating that account number and routing number information may typically be required for completing bank account transactions in the United States using a USD currency type. Bank account data parameters of a given country block entry stored in validation database **120** may additionally correspond to a currency type associated with a supplier bank account transaction (e.g., as indicated by a currency identifier). In the example described above, bank account data parameters of the country block entry for the United States may additionally include, or be updated to include, wire routing number and bank code parameters if the bank account transaction requires a EUR currency type rather than USD. In one embodiment, each bank account data parameter stored in validation database **120** may be or include a regular expression used to define a respective search pattern for supplier data received via the spend management interface **110**. That is, the validation subsystem **130** may use the search pattern corresponding to the regular expression for each bank account data parameter to validate supplier data.

In one embodiment, the validation database **120** may be represented by one or more versions (e.g., version 1.0, 2.0, and the like). Here, bank account data parameters of a given country block entry stored in validation database **120** may additionally correspond to a respective version of the validation database **120** that the validation subsystem **130** uses to validate supplier data. Each respective version of the validation database **120** may correspond to a respective version of the spend management system **100**, and/or spend management interface **110** therein, and may provide different degrees of supplier data validation. For example, a legacy version of the validation database **120** in use for 10 years or more may provide less strict supplier data validation than a newer version of the validation database **120** in use for less than 10 years. That is, each bank account parameter stored in previous versions of the validation database **120** may be or include a regular expression used to define a respective search pattern for supplier data that may result in a more lenient search than regular expressions comprising bank account data parameters stored in newer versions of the validation database **120**. In one embodiment, the spend management system **100** may allow buyer computers **160** to elect, or otherwise opt-in, to using an updated version of the validation database **120** for performing advanced supplier data validation or to remain using previous versions of the validation database **120** for performing less strict supplier data validations. In one embodiment, the validation database **120** may be or include a relational database in which bank account data parameters are digitally stored within one or more country block entries as described above. In other embodiments, the validation database **120** may be or include a centralized database, distributed database, commercial database, operational database, and/or any other database management system suitable for storing invoices generated by the supplier computers **150**.

In one embodiment, the validation subsystem **130** may comprise a system, device, or apparatus executing stored program instructions that are programmed to validate supplier data provided by supplier computers **150**. In one embodiment, the validation subsystem **130** may be programmed to access the spend management interface **110** of the spend management system **100** such that the validation

subsystem **130** may receive supplier data from supplier computers **150** via user interface widgets electronically presented within the spend management interface **110**. Specifically, the validation subsystem **130** may receive a country identifier indicating a country in which a supplier bank account is located via a country widget electronically presented within the spend management interface **110**. The validation subsystem **130** may additionally receive a currency identifier indicating a currency type associated with a supplier bank account transaction via a currency widget electronically presented within the spend management interface **110**. In another embodiment, the spend management system **100** may be programmed to extract supplier data from the spend management interface **110** such that the country identifier and/or currency identifier may be accessed directly by the validation subsystem **130**.

In one embodiment, the validation subsystem **130** may be programmed to identify one or more parameters of multiple bank account data parameters stored in validation database **120**. In particular, the validation subsystem **130** may select a country block entry stored in the validation database **120** based on the country identifier received from the supplier computer **150** (e.g., via the country widget electronically presented within the spend management interface **110**) and may identify parameters included in the country block entry accordingly. In one embodiment, the validation subsystem **130** may identify parameters included in the country block entry once the country block entry has been selected. For example, the validation subsystem **130** may receive a country identifier indicating the United States, select the country block entry corresponding to the United States, and identify account number and routing number parameters included in the country block entry. In another embodiment, the validation subsystem **130** may identify parameters included in the country block entry once the country block entry has been selected and a currency type has been identified. Here, the validation subsystem **130** may identify a currency type based on the currency identifier received from the supplier computer **150** (e.g., via the currency widget electronically presented within the spend management interface **110**). In the example described above, the validation subsystem **130** may receive the country identifier indicating the United States, receive a currency identifier indicating a EUR currency type, select the country block entry corresponding to the United States, identify the EUR currency type indicated by the currency identifier, and identify the wire routing number and bank code parameters included in the country block entry based on the currency identifier. In one embodiment, the one or more parameters identified by the validation subsystem **130** may include a bank code, a routing number, and an account number. In other embodiments, the one or more parameters may additionally include a wire routing number, a Society for Worldwide Interbank Financial Telecommunication (SWIFT) code, an IBAN, a sort code, a Bank State Branch (BSB) code, an Indian Financial System Code (IFSC), a transit number, an institution number, and/or any suitable combination of parameters stored in validation database **120**.

In one embodiment, the validation subsystem **130** may be programmed to generate one or more additional user interface widgets, or "customized user interface widgets," based on the one or more parameters identified by the validation subsystem **130**. In particular, the validation subsystem **130** may identify one or more parameters included in country block entries stored in the validation database **120** as described above and may generate customized user interface widgets based on the one or more parameters. The custom-

ized user interface widgets generated by the validation subsystem **130** may be associated with all data values of supplier data necessary to correctly address a supplier bank account to complete a supplier bank account transaction. For example, upon identifying wire routing number and bank code parameters associated with a supplier bank account transaction in the United States using a EUR currency type, the validation subsystem **130** may generate customized user interface widgets configured to receive wire routing number and bank code information from a supplier computer **150**. In this example, the wire routing number and bank code information may be subsequently used by a buyer computer **160** for correctly addressing the supplier bank account using the specified currency type to complete the supplier bank account transaction. In one embodiment, each customized user interface widget generated by the validation subsystem **130** may be comprised of a data field configured to receive supplier data having a proper syntax, format, and/or length corresponding to the parameters on which the customized user interface widget is based. For example, a customized user interface widget generated by the validation subsystem **130** to receive account number information from a supplier computer **150** for a supplier bank account transaction in the United States using a USD currency type may be configured to receive up to 17 numeric digits from the supplier computer **150**.

In one embodiment, the validation subsystem **130** may be programmed to electronically update the spend management interface **110** to present the one or more customized user interface widgets to supplier computers **150**. Specifically, the validation subsystem **130** may electronically update, or otherwise modify, the software framework comprising the spend management interface **110**, and/or one or more graphical controls presented therein, to present the one or more customized user interface widgets to supplier computers **150**. That is, the spend management interface **110** may be electronically updated to present the one or more customized user interface widgets rather than the one or more user interface widgets initially presented to the supplier computers **150** as described above with respect to the spend management interface **110**. Once the spend management interface **110** has been electronically updated, the supplier computers **150** may manually enter supplier data into the spend management system **100** using the one or more customized user interface widgets.

In one embodiment, the validation subsystem **130** may programmatically validate the supplier data received via the one or more customized user interface widgets. As described above with respect to the validation database **120**, each bank account data parameter stored in the validation database **120** may be or include a regular expression used to define a respective search pattern for supplier data received via the spend management interface **110**. That is, the validation subsystem **130** may use the search pattern corresponding to the regular expression for each bank account data parameter to validate supplier data with respect to proper syntax, format, and/or length. Upon receiving supplier data via the one or more customized user interface widgets, the validation subsystem **130** may use the regular expressions corresponding to the one or more parameters upon which the one or more customized user interface widgets are based to programmatically validate the supplier data. In particular, the validation subsystem **130** may use the search pattern for each of the regular expressions corresponding to the one or more parameters to programmatically match the search pattern to respective target strings comprising the supplier data. In one embodiment, the validation subsystem **130** may

programmatically validate supplier data using Basic Regular Expressions (BRE). In other embodiments, the validation subsystem **130** may programmatically validate supplier data using Extended Regular Expressions (ERE), Simple Regular Expressions (SRE), and/or any combination of regular expressions suitable for validating supplier data.

In one embodiment, the validation subsystem **130** may programmatically verify the supplier data received via the one or more customized user interface widgets. That is, in addition to validating supplier data with respect to proper syntax, format, and/or length as described above, the validation subsystem **130** may verify that a supplier bank account is a valid account. Upon receiving a supplier routing number via a customized user interface widget electronically presented within the spend management interface **110**, the validation subsystem **130** may access a list, or directory, of valid routing numbers (e.g., American Bankers Association (ABA) directory) and may programmatically match the supplier routing number to a valid routing number included in the list of valid routing numbers. In this way, the validation subsystem **130** may, both, validate supplier data provided by supplier computers **150** with respect to proper syntax, format, and/or length and verify that the validated supplier data is accessible to buyers for completing bank account transactions. In one embodiment, the list, or directory, of valid routing numbers may be stored in the validation database **120**.

Network **140** may be implemented by any medium or mechanism that provides for the exchange of data between the various elements of FIG. 1. Examples of network **140** include, without limitation, a cellular network, communicatively coupled with a data connection to the computing devices over a cellular antenna, one or more Local Area Networks (LANs), one or more Wide Area Networks (WANs), one or more Ethernets or the Internet, or one or more terrestrial, satellite or wireless links, or a combination thereof. For purposes of illustrating a clear example, network **140** is shown as a single element but in practice, network **140** may comprise one or more local area networks, wide area networks, and/or internetworks. The various elements of FIG. 1 may also have direct (e.g., wired or wireless) communications links, depending upon a particular implementation.

2.2 User Interface Example

FIG. 2A, FIG. 2B, FIG. 2C illustrate an example spend management interface of a spend management system, according to various embodiments. In this context, "supplier data" refers to electronic information, or a digital dataset, that is digitally stored in an electronic storage device and managed programmatically using the techniques that are described herein in other sections.

Embodiments may use graphical user interfaces that buyer computers or supplier computers render based on dynamically generated markup language code that is locally executed at the buyer computers or supplier computers. In the embodiment illustrated in FIG. 2A, the spend management interface **110** has been rendered at a buyer computer based on receiving markup language code from a spend management system instance, and includes a country widget **200** configured to receive a country identifier **201** from a supplier computer **150** (shown in FIG. 1), a currency widget (not shown in figure) configured to receive a currency identifier (not shown in figure) from a supplier computer **150**, and multiple user interface widgets **204** electronically presented within the spend management interface **110**. As

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shown in FIG. 2A, the user interface widgets **204** may include a “Beneficiary Name” widget, a “Bank Name” widget, an “Account Number” widget, a “Confirm Account Number” widget, a “Routing Number” widget, a “SWIFT/BIC Code” widget, and a “Bank Account Type” widget. It is noted that other embodiments of the spend management interface **110** may include additional, fewer, and/or different user interface widgets than those illustrated in FIG. 2A.

In one embodiment, a supplier computer **150** may access the spend management interface **110** and a supplier may manually enter a country identifier **201** into the spend management system **100** via the country widget **200**. This is shown in FIG. 2A where a supplier using a supplier computer **150** has selected “United States” from a drop-down menu listing multiple countries alphabetically. In other embodiments, suppliers may manually enter country identifiers **201** into the spend management system **100** via a country widget **200** having a plain text field, hover menu, pull-down list, picklist, and/or any combination of widgets suitable for receiving a country identifier **201** from supplier computers **150**.

FIG. 2B illustrates the example spend management interface **110** shown in FIG. 2A. In the example shown in FIG. 2B, the spend management interface **110** is in a state where the supplier has selected the country identifier **201** from the drop-down menu shown in FIG. 2A. Specifically, the currency widget **202** and currency identifier **203** are visible within the spend management interface **110** in addition to hint icons **208** and suggestion **210**. As shown in FIG. 2B, the currency identifier **203** in the currency widget **202** indicates a USD currency type. In one embodiment, the currency identifier **203** may autofill the currency widget **202** to a default currency type representative of the country in which the supplier bank account is located (i.e., as indicated by the country identifier **201**). For example, upon receiving the country identifier **201** from a supplier computer **150**, the validation subsystem **130** may select a country block entry stored in the validation database **120** based on the country identifier **201**, identify a parameter indicating a currency type included in the country block entry, and autofill the currency widget **202** with the currency type indicated by the parameter. In another embodiment, a supplier may access a list of available currency types via the spend management interface **110**. For example, a supplier may access a drop-down menu listing multiple currency types alphabetically, similar to the drop-down menu shown in FIG. 2A. In other embodiments, suppliers may manually enter currency identifiers **203** into the spend management system **100** via a currency widget **202** having a plain text field, hover menu, pull-down list, picklist, and/or any combination of widgets suitable for receiving a currency identifier **203** from supplier computers **150**.

As shown in FIG. 2B, the spend management interface **110** may include one or more hint icons **208** disposed proximate to one or more user interface widgets **204**. Each hint icon **208** electronically presented within the spend management interface **110** may be programmed to provide suppliers with suggestions **210** regarding an expected format of supplier data (e.g., with respect to syntax, length, and the like) to be manually entered by suppliers. Each suggestion **210** provided to suppliers using hint icons **208** may correspond to the user interface widget next to which the hint icon **208** is presented. In the example shown in FIG. 2B, the suggestion **210** “Please enter the Account Number (up to 17 digits)” is presented proximate to the “Account Number” widget, indicating that the suggestion **210** corresponds to a supplier account number. In one embodiment, a supplier

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computer **150** may initiate a request to receive suggestions **210**, or a “hint request,” by simply clicking on, hovering over, or otherwise engaging with, hint icons **208** electronically presented within the spend management interface **110**. In response to receiving the hint request via hint icons **208**, the validation subsystem **130** may access the validation database **120** and identify a suggestion **210** based on the user interface widget next to which the hint icon **208** is presented. For example, suggestions **210** may be stored in validation database **120** using a string format, where each suggestion **210** may be indexed based on the user interface widget to which the suggestion **210** corresponds. Once the suggestion **210** has been identified, the validation subsystem **130** may electronically present the suggestion **210** to the supplier computer **150** via the spend management interface **110** as shown in FIG. 2B.

FIG. 2C illustrates the example spend management interface **110** shown in FIG. 2A and FIG. 2B. In the example shown in FIG. 2C, the spend management interface **110** is in a state where the supplier has selected a different currency identifier **203** than that shown in FIG. 2B. Specifically, the supplier has changed the currency identifier **203** from a USD currency type to a EUR currency type. In response to a change in currency type, the validation subsystem **130** may electronically update, or otherwise modify, the software framework comprising the spend management interface **110** to present one or more customized user interface widgets based on one or more parameters stored in validation database **120** as described above with respect to FIG. 1. This is shown in FIG. 2C where the spend management interface **110** includes a “Routing (Bank Code) Number” widget **214** rather than the “Routing Number” widget **206** shown in FIG. 2B. Once the spend management interface **110** has been electronically updated, the supplier may manually enter supplier data into the spend management system **100** using the one or more customized user interface widgets. In this way, the validation subsystem **130** may ensure that the supplier data received via the spend management interface **110** corresponds to the appropriate country identifier **201** and currency identifier **203** for receiving payments. Presenting buyers with this accurate supplier data may prevent, both, failures to accurately integrate supplier data across multiple spend management systems accessed by buyers and costly bounced payments from banks due to buyers attempting to inaccurate supplier data.

2.3 Supplier Data Validation Example

FIG. 3A and FIG. 3B illustrate an example spend management interface that includes an error notification, according to various embodiments. In this context, “error notification” refers to electronic information, or a digital dataset, that is digitally stored in an electronic storage device and managed programmatically using the techniques that are described herein in other sections. In one embodiment, the validation subsystem **130** may access the validation database **120** and identify error notifications based on one or more user interface widgets **204** that have received invalid supplier data. For example, error notifications may be stored in validation database **120** using a string format, where each error notification may be indexed based on the user interface widget to which the error notification corresponds.

FIG. 3A illustrates the example spend management interface **110** shown in FIG. 2A, FIG. 2B, FIG. 2C. As shown in FIG. 3A, the country identifier **201** in the country widget **200** indicates the United States and the currency identifier **203** in the currency widget **202** indicates a USD currency type.

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Because routing number information may typically be required for completing bank account transactions in the United States using a USD currency type, the validation subsystem 130 has electronically updated the spend management interface 110 to present one or more customized user interface widgets that include a "Routing Number" widget 300. In the example shown in FIG. 3A, the spend management interface 110 is in a state where a supplier using a supplier computer 150 has attempted to manually enter a routing number into the "Routing Number" widget 300. However, an error notification 302 has been electronically presented within the spend management interface 110. Specifically, the error notification 302 "must be 9 digits" has been presented proximate to the "Routing Number" widget 300, indicating that the routing number manually entered by the supplier is invalid. To electronically present the error notification 302 within the spend management interface 110, the validation subsystem 130 may first attempt to validate the routing number manually entered into the "Routing Number" widget 300 by the supplier. In particular, the validation subsystem 130 may use the regular expression corresponding to the routing number parameter included in the country block entry corresponding to the country identifier 201 (i.e., the United States) and currency identifier 203 (i.e., USD currency type) to programmatically validate the supplier data. Upon determining that the supplier data received from the supplier computer 150 via the "Routing Number" widget 300 is invalid, the validation subsystem 130 may then electronically present the error notification 302 to the supplier computer 150 indicating that the supplier data received is invalid.

FIG. 3B illustrates the example spend management interface 110 shown in FIG. 2A, FIG. 2B, FIG. 2C, FIG. 3A. As shown in FIG. 3B, the country identifier 201 in the country widget 200 indicates Australia and the currency identifier 203 in the currency widget 202 indicates an Australian dollar (AUD) currency type. Because BSB information may typically be required for completing bank account transactions in Australia using an AUD currency type, the validation subsystem 130 has electronically updated the spend management interface 110 to present one or more customized user interface widgets that include a "BSB" widget 304. In the example shown in FIG. 3B, the spend management interface 110 is in a state where a supplier using a supplier computer 150 has attempted to manually enter a BSB code into the "BSB" widget 304. However, an error notification 306 has been electronically presented within the spend management interface 110. In particular, the error notification 306 "must be 6 digits" has been presented proximate to the "BSB" widget 304, indicating that the BSB code manually entered by the supplier is invalid. To electronically present the error notification 306 within the spend management interface 110, the validation subsystem 130 may first attempt to validate the BSB code manually entered into the "BSB" widget 304 by the supplier. In particular, the validation subsystem 130 may use the regular expression corresponding to the BSB parameter included in the country block entry corresponding to the country identifier 201 (i.e., Australia) and the currency identifier 203 (i.e., AUD currency type) to programmatically validate the supplier data. Upon determining that the supplier data received from the supplier computer 150 via the "BSB" widget 304 is invalid, the validation subsystem 130 may then electronically present the error notification 306 to the supplier computer 150 indicating that the supplier data received is invalid.

3. Functional Overview

FIG. 4 illustrates a flow diagram of an algorithm that may be programmed to implement an embodiment. FIG. 4, and

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the other drawing figures and all of the description and claims in this disclosure, are intended to present, disclose and claim a technical system and technical methods in which specially programmed computers, using a special-purpose distributed computer system design, execute functions that have not been available before to provide a practical application of supplier data validation using a validation subsystem. In this manner, the disclosure presents a technical solution to a technical problem, and any interpretation of the disclosure or claims to cover any judicial exception to patent eligibility, such as an abstract idea, mental process, method of organizing human activity or mathematical algorithm, has no support in this disclosure and is erroneous.

As an overview, the process of FIG. 4 is generally directed to a method for validating supplier data. The method may begin at step 410, where the process may be programmed to access a spend management interface of a spend management system. The spend management interface may be programmed to receive supplier data from a supplier computer via one or more user interface widgets of the spend management interface. For example, the validation subsystem 130 shown in FIG. 1 may access the spend management interface 110 that includes user interface widgets 204 as shown in FIG. 2A.

At step 420, the process may be programmed to receive a country identifier from the supplier computer via a country widget of the one or more user interface widgets. The country identifier may indicate a country in which a supplier bank account is located. For example, the validation subsystem 130 may receive the country identifier 201 via the country widget 200 indicating the United States as shown in FIG. 2B.

At step 430, the process may be programmed to receive a currency identifier from the supplier computer via a currency widget of the one or more user interface widgets. The currency identifier may indicate a currency type associated with a supplier bank account transaction. For example, the validation subsystem 130 may receive the currency identifier 203 via the currency widget 202 indicating a USD currency type as shown in FIG. 2B.

At step 440, the process may be programmed to identify one or more parameters of multiple bank account data parameters stored in a validation database of the spend management system. The one or more parameters may be associated with the country identifier and the currency identifier. For example, the validation subsystem 130 may identify one or more parameters of multiple bank account data parameters stored in validation database shown in FIG. 1. The one or more parameters may be associated with the country identifier 201 and the currency identifier 203 shown in FIG. 2A. In one embodiment, the one or more parameters identified by the validation subsystem 130 may include a bank code, a routing number, and an account number. In other embodiments, the one or more parameters may additionally include a wire routing number, a Society for Worldwide Interbank Financial Telecommunication (SWIFT) code, an IBAN, a sort code, a Bank State Branch (BSB) code, an Indian Financial System Code (IFSC), a transit number, an institution number, and/or any suitable combination of parameters stored in validation database 120 shown in FIG. 1.

At steps 450 and 460, the process may be programmed to generate one or more customized user interface widgets based on the one or more parameters and to electronically update the spend management interface to present the one or more customized user interface widgets to the supplier computer, respectively. The one or more customized user

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interface widgets may be associated with all data values of the supplier data necessary to correctly address the supplier bank account for the supplier bank account transaction. For example, the validation subsystem **130** may generate the “Routing Number” widget **206** shown in FIG. 2B to receive routing number information necessary for completing bank account transactions in the United States using a USD currency type. In another example, the validation subsystem **130** may generate the “Routing (Bank Code) Number” widget **214** shown in FIG. 2C to receive wire routing number and bank code information necessary for completing bank account transactions in the United States using a EUR currency type.

At steps **470** and **480**, the process may be programmed to receive the supplier data from the supplier computer via the one or more customized user interface widgets and to programmatically validate the supplier data. For example, the validation subsystem **130** may receive routing number information via the “Routing Number” widget **300** shown in FIG. 3A. The validation subsystem **130** may use the regular expression corresponding to the routing number parameter included in the country block entry corresponding to the country identifier **201** (i.e., the United States) and currency identifier **203** (i.e., USD currency type) to programmatically validate the supplier data as described above with respect to FIG. 3A. In another example, the validation subsystem **130** may receive BSB information via the “BSB” widget **304** shown in FIG. 3B. The validation subsystem **130** may use the regular expression corresponding to the BSB parameter included in the country block entry corresponding to the country identifier **201** (i.e., Australia) and the currency identifier **203** (i.e., AUD currency type) to programmatically validate the supplier data as described above with respect to FIG. 3B.

Using these approaches, embodiments provide the benefit of a technical solution to the technical problems identified in the Background in at least the following manner. First, electronically presenting one or more user interface widgets within the user interface of a spend management system that represent specific syntax requirements corresponding to the country in which a supplier bank account is located and/or the given currency type associated with a supplier bank account transaction may encourage suppliers to manually enter accurate supplier data into the spend management system. Thus, the accurate supplier data may be more readily integrated across multiple spend management systems accessed by buyers (e.g., enterprise resource planning (ERP) platforms) for making payments to suppliers. This may greatly reduce instances of costly bounced payments from banks due to buyers attempting to use inaccurate supplier data for supplier bank account transactions. Second, the security and resistance to fraud of a spend management system, e-procurement system, or invoice processing system are improved because the automatic stored program techniques described herein will reduce the quantity of inaccurate supplier data that a supplier can generate, thereby deterring buyers from seeking new mechanisms for supplier data validation (e.g., third-party software) which may expose sensitive data and present opportunities for fraud and/or disruptions in supplier bank account transactions. Third, reductions in the number of CPU cycles, memory, storage, network bandwidth, and other resources are achieved because the automatic stored program techniques described herein may prevent the storage and incorrect processing of inaccurate supplier data that has been created in a spend management system, e-procurement system, or invoice processing system erroneously or intentionally.

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4. Implementation Example—Hardware Overview

FIG. 5 is a block diagram that illustrates an example computer system with which an embodiment may be implemented. In the example of FIG. 5, a computer system and instructions for implementing the disclosed technologies in hardware, software, or a combination of hardware and software, are represented schematically, for example as boxes and circles, at the same level of detail that is commonly used by persons of ordinary skill in the art to which this disclosure pertains for communicating about computer architecture and computer systems implementations.

Computer system includes an input/output (I/O) subsystem **502** which may include a bus and/or other communication mechanism(s) for communicating information and/or instructions between the components of the computer system over electronic signal paths. The I/O subsystem **502** may include an I/O controller, a memory controller and at least one I/O port. The electronic signal paths are represented schematically in the drawings, for example as lines, unidirectional arrows, or bidirectional arrows.

At least one hardware processor **504** is coupled to I/O subsystem **502** for processing information and instructions. Hardware processor **504** may include, for example, a general-purpose microprocessor or microcontroller and/or a special-purpose microprocessor such as an embedded system or a graphics processing unit (GPU) or a digital signal processor or ARM processor. Processor **504** may comprise an integrated arithmetic logic unit (ALU) or may be coupled to a separate ALU.

Computer system includes one or more units of memory **506**, such as a main memory, which is coupled to I/O subsystem **502** for electronically digitally storing data and instructions to be executed by processor **504**. Memory **506** may include volatile memory such as various forms of random-access memory (RAM) or other dynamic storage device. Memory **506** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **504**. Such instructions, when stored in non-transitory computer-readable storage media accessible to processor **504**, can render computer system into a special-purpose machine that is customized to perform the operations specified in the instructions.

Computer system further includes non-volatile memory such as read only memory (ROM) **508** or other static storage device coupled to I/O subsystem **502** for storing information and instructions for processor **504**. The ROM **508** may include various forms of programmable ROM (PROM) such as erasable PROM (EPROM) or electrically erasable PROM (EEPROM). A unit of persistent storage **510** may include various forms of non-volatile RAM (NVRAM), such as FLASH memory, or solid-state storage, magnetic disk or optical disk such as CD-ROM or DVD-ROM and may be coupled to I/O subsystem **502** for storing information and instructions. Storage **510** is an example of a non-transitory computer-readable medium that may be used to store instructions and data which when executed by the processor **504** cause performing computer-implemented methods to execute the techniques herein.

The instructions in memory **506**, ROM **508** or storage **510** may comprise one or more sets of instructions that are organized as modules, methods, objects, functions, routines, or calls. The instructions may be organized as one or more computer programs, operating system services, or application programs including mobile apps. The instructions may comprise an operating system and/or system software; one

or more libraries to support multimedia, programming or other functions; data protocol instructions or stacks to implement TCP/IP, HTTP or other communication protocols; file format processing instructions to parse or render files coded using HTML, XML, JPEG, MPEG or PNG; user interface instructions to render or interpret commands for a graphical user interface (GUI), command-line interface or text user interface; application software such as an office suite, internet access applications, design and manufacturing applications, graphics applications, audio applications, software engineering applications, educational applications, games or miscellaneous applications. The instructions may implement a web server, web application server or web client. The instructions may be organized as a presentation layer, application layer and data storage layer such as a relational database system using structured query language (SQL) or no SQL, an object store, a graph database, a flat file system or other data storage.

Computer system may be coupled via I/O subsystem **502** to at least one output device **512**. In one embodiment, output device **512** is a digital computer display. Examples of a display that may be used in various embodiments include a touch screen display or a light-emitting diode (LED) display or a liquid crystal display (LCD) or an e-paper display. Computer system may include other type(s) of output devices **512**, alternatively or in addition to a display device. Examples of other output devices **512** include printers, ticket printers, plotters, projectors, sound cards or video cards, speakers, buzzers or piezoelectric devices or other audible devices, lamps or LED or LCD indicators, haptic devices, actuators or servos.

At least one input device **514** is coupled to I/O subsystem **502** for communicating signals, data, command selections or gestures to processor **504**. Examples of input devices **514** include touch screens, microphones, still and video digital cameras, alphanumeric and other keys, keypads, keyboards, graphics tablets, image scanners, joysticks, clocks, switches, buttons, dials, slides, and/or various types of sensors such as force sensors, motion sensors, heat sensors, accelerometers, gyroscopes, and inertial measurement unit (IMU) sensors and/or various types of transceivers such as wireless, such as cellular or Wi-Fi, radio frequency (RF) or infrared (IR) transceivers and Global Positioning System (GPS) transceivers.

Another type of input device is a control device **516**, which may perform cursor control or other automated control functions such as navigation in a graphical interface on a display screen, alternatively or in addition to input functions. Control device **516** may be a touchpad, a mouse, a trackball, or cursor direction keys for communicating direction information and command selections to processor **504** and for controlling cursor movement on output device **512**. The input device may have at least two degrees of freedom in two axes, a first axis (e.g., x) and a second axis (e.g., y), that allows the device to specify positions in a plane. Another type of input device is a wired, wireless, or optical control device such as a joystick, wand, console, steering wheel, pedal, gearshift mechanism or other type of control device. An input device **514** may include a combination of multiple different input devices, such as a video camera and a depth sensor.

In another embodiment, computer system may comprise an internet of things (IoT) device in which one or more of the output device **512**, input device **514**, and control device **516** are omitted. Or, in such an embodiment, the input device **514** may comprise one or more cameras, motion detectors, thermometers, microphones, seismic detectors, other sen-

sors or detectors, measurement devices or encoders and the output device **512** may comprise a special-purpose display such as a single-line LED or LCD display, one or more indicators, a display panel, a meter, a valve, a solenoid, an actuator or a servo.

When computer system is a mobile computing device, input device **514** may comprise a global positioning system (GPS) receiver coupled to a GPS module that is capable of triangulating to a plurality of GPS satellites, determining and generating geo-location or position data such as latitude-longitude values for a geophysical location of the computer system. Output device **512** may include hardware, software, firmware and interfaces for generating position reporting packets, notifications, pulse or heartbeat signals, or other recurring data transmissions that specify a position of the computer system, alone or in combination with other application-specific data, directed toward host **524** or server **530**.

Computer system may implement the techniques described herein using customized hard-wired logic, at least one ASIC or FPGA, firmware and/or program instructions or logic which when loaded and used or executed in combination with the computer system causes or programs the computer system to operate as a special-purpose machine. According to one embodiment, the techniques herein are performed by computer system in response to processor **504** executing at least one sequence of at least one instruction contained in main memory **506**. Such instructions may be read into main memory **506** from another storage medium, such as storage **510**. Execution of the sequences of instructions contained in main memory **506** causes processor **504** to perform the process steps described herein. In alternative embodiments, hard-wired circuitry may be used in place of or in combination with software instructions.

The term "storage media" as used herein refers to any non-transitory media that store data and/or instructions that cause a machine to operation in a specific fashion. Such storage media may comprise non-volatile media and/or volatile media. Non-volatile media includes, for example, optical or magnetic disks, such as storage **510**. Volatile media includes dynamic memory, such as memory **506**. Common forms of storage media include, for example, a hard disk, solid state drive, flash drive, magnetic data storage medium, any optical or physical data storage medium, memory chip, or the like.

Storage media is distinct from but may be used in conjunction with transmission media. Transmission media participates in transferring information between storage media. For example, transmission media includes coaxial cables, copper wire and fiber optics, including the wires that comprise a bus of I/O subsystem **502**. Transmission media can also take the form of acoustic or light waves, such as those generated during radio-wave and infrared data communications.

Various forms of media may be involved in carrying at least one sequence of at least one instruction to processor **504** for execution. For example, the instructions may initially be carried on a magnetic disk or solid-state drive of a remote computer. The remote computer can load the instructions into its dynamic memory and send the instructions over a communication link such as a fiber optic or coaxial cable or telephone line using a modem. A modem or router local to computer system can receive the data on the communication link and convert the data to a format that can be read by computer system. For instance, a receiver such as a radio frequency antenna or an infrared detector can receive the data carried in a wireless or optical signal and appropriate circuitry can provide the data to I/O subsystem **502** such as

place the data on a bus. I/O subsystem **502** carries the data to memory **506**, from which processor **504** retrieves and executes the instructions. The instructions received by memory **506** may optionally be stored on storage **510** either before or after execution by processor **504**.

Computer system also includes a communication interface **518** coupled to network **522**. Communication interface **518** provides a two-way data communication coupling to network link(s) **520** that are directly or indirectly connected to at least one communication networks, such as a network **522** or a public or private cloud on the Internet. For example, communication interface **518** may be an Ethernet networking interface, integrated-services digital network (ISDN) card, cable modem, satellite modem, or a modem to provide a data communication connection to a corresponding type of communications line, for example an Ethernet cable or a metal cable of any kind or a fiber-optic line or a telephone line. Network **522** broadly represents a local area network (LAN), wide-area network (WAN), campus network, inter-network or any combination thereof. Communication interface **518** may comprise a LAN card to provide a data communication connection to a compatible LAN, or a cellular radiotelephone interface that is wired to send or receive cellular data according to cellular radiotelephone wireless networking standards, or a satellite radio interface that is wired to send or receive digital data according to satellite wireless networking standards. In any such implementation, communication interface **518** sends and receives electrical, electromagnetic or optical signals over signal paths that carry digital data streams representing various types of information.

Network link **520** typically provides electrical, electromagnetic, or optical data communication directly or through at least one network to other data devices, using, for example, satellite, cellular, Wi-Fi, or BLUETOOTH technology. For example, network link **520** may provide a connection through a network **522** to a host computer **524**.

Furthermore, network link **520** may provide a connection through network **522** or to other computing devices via internetworking devices and/or computers that are operated by an Internet Service Provider (ISP) **526**. ISP **526** provides data communication services through a world-wide packet data communication network represented as internet **528**. A server computer **530** may be coupled to internet **528**. Server **530** broadly represents any computer, data center, virtual machine or virtual computing instance with or without a hypervisor, or computer executing a containerized program system such as DOCKER or KUBERNETES. Server **530** may represent an electronic digital service that is implemented using more than one computer or instance and that is accessed and used by transmitting web services requests, uniform resource locator (URL) strings with parameters in HTTP payloads, API calls, app services calls, or other service calls. Computer system and server **530** may form elements of a distributed computing system that includes other computers, a processing cluster, server farm or other organization of computers that cooperate to perform tasks or execute applications or services. Server **530** may comprise one or more sets of instructions that are organized as modules, methods, objects, functions, routines, or calls. The instructions may be organized as one or more computer programs, operating system services, or application programs including mobile apps. The instructions may comprise an operating system and/or system software; one or more libraries to support multimedia, programming or other functions; data protocol instructions or stacks to implement TCP/IP, HTTP or other communication protocols; file for-

mat processing instructions to parse or render files coded using HTML, XML, JPEG, MPEG or PNG; user interface instructions to render or interpret commands for a graphical user interface (GUI), command-line interface or text user interface; application software such as an office suite, internet access applications, design and manufacturing applications, graphics applications, audio applications, software engineering applications, educational applications, games or miscellaneous applications. Server **530** may comprise a web application server that hosts a presentation layer, application layer and data storage layer such as a relational database system using structured query language (SQL) or no SQL, an object store, a graph database, a flat file system or other data storage.

Computer system can send messages and receive data and instructions, including program code, through the network(s), network link **520** and communication interface **518**. In the Internet example, a server **530** might transmit a requested code for an application program through internet **528**, ISP **526**, local network **522** and communication interface **518**. The received code may be executed by processor **504** as it is received, and/or stored in storage **510**, or other non-volatile storage for later execution.

The execution of instructions as described in this section may implement a process in the form of an instance of a computer program that is being executed and consisting of program code and its current activity. Depending on the operating system (OS), a process may be made up of multiple threads of execution that execute instructions concurrently. In this context, a computer program is a passive collection of instructions, while a process may be the actual execution of those instructions. Several processes may be associated with the same program; for example, opening up several instances of the same program often means more than one process is being executed. Multitasking may be implemented to allow multiple processes to share processor **504**. While each processor **504** or core of the processor executes a single task at a time, computer system may be programmed to implement multitasking to allow each processor to switch between tasks that are being executed without having to wait for each task to finish. In an embodiment, switches may be performed when tasks perform input/output operations, when a task indicates that it can be switched, or on hardware interrupts. Time-sharing may be implemented to allow fast response for interactive user applications by rapidly performing context switches to provide the appearance of concurrent execution of multiple processes simultaneously. In an embodiment, for security and reliability, an operating system may prevent direct communication between independent processes, providing strictly mediated and controlled inter-process communication functionality.

5. Benefits; Extensions and Alternatives

Embodiments provide the benefit of a technical solution to the technical problems identified in the Background in at least the following manner. First, electronically presenting one or more user interface widgets within the user interface of a spend management system that represent specific syntax requirements corresponding to the country in which a supplier bank account is located and/or the given currency type associated with a supplier bank account transaction may encourage suppliers to manually enter accurate supplier data into the spend management system. Thus, the accurate supplier data may be more readily integrated across multiple spend management systems accessed by buyers (e.g., enter-

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prise resource planning (ERP) platforms) for making payments to suppliers. This may greatly reduce instances of costly bounced payments from banks due to buyers attempting to use inaccurate supplier data for supplier bank account transactions. Second, the security and resistance to fraud of a spend management system, e-procurement system, or invoice processing system are improved because the automatic stored program techniques described herein will reduce the quantity of inaccurate supplier data that a supplier can generate, thereby deterring buyers from seeking new mechanisms for supplier data validation (e.g., third-party software) which may expose sensitive data and present opportunities for fraud and/or disruptions in supplier bank account transactions. Third, reductions in the number of CPU cycles, memory, storage, network bandwidth, and other resources are achieved because the automatic stored program techniques described herein may prevent the storage and incorrect processing of inaccurate supplier data that has been created in a spend management system, e-procurement system, or invoice processing system erroneously or intentionally.

In the foregoing specification, embodiments of the invention have been described with reference to numerous specific details that may vary from implementation to implementation. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. The sole and exclusive indicator of the scope of the invention, and what is intended by the applicants to be the scope of the invention, is the literal and equivalent scope of the set of claims that issue from this application, in the specific form in which such claims issue, including any subsequent correction.

What is claimed is:

1. A method by a spend management system comprising a validation database, the method comprising:
 - storing, by the spend management system, a plurality of country block entries in the validation database, wherein each country block entry of the plurality of country block entries includes a bank code parameter and a wire routing number and each of the bank code parameters comprises a regular expression with an associated search pattern;
 - generating and displaying, by the spend management system, a spend management interface comprising a plurality of user interface widgets including a country widget and a currency widget;
 - receiving, by the spend management interface, via the country widget, a country identifier from a supplier computer;
 - receiving, by the spend management interface, via the currency widget, a currency identifier from the supplier computer;
 - selecting, by the spend management system, based on the country identifier, a country block entry from the plurality of country block entries and selecting, by the spend management system, a bank code parameter and wire routing number from the selected country block entry;
 - autofilling, by the spend management interface, one or more user interface widgets among the plurality of user interface widgets with the wire routing number and bank code parameters and generating, by the spend management interface, a user interface comprising the autofilled widgets;
 - displaying, by the spend management interface, the user interface;

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receiving, by the displayed user interface, supplier data via the autofilled widgets;

validating, by the spend management interface, based on the search pattern associated with the regular expression of the selected bank code parameter and wire routing number, the supplier data received via the autofilled widgets; and

transmitting, by the spend management interface, the validated supplier data to a plurality of spend management systems.

2. The method of claim 1, further comprising:

identifying, by the spend management system, a currency type, wherein the currency type is associated with the currency identifier; and

determining, by the spend management system, each of the bank code parameter and wire routing parameter in a corresponding country block entry based on the currency identifier.

3. The method of claim 1, further comprising:

displaying, by the spend management interface, one or more hint icons and receiving, by the spend management interface, a hint request from the supplier computer, via the displayed one or more icons;

electronically presenting, by the spend management interface, a suggestion associated with the one or more hint icons.

4. The method of claim 1, each country block entry of the plurality of country block entries further including an account number, the method further comprising:

storing, by the validation database, a list of valid routing numbers;

verifying, by the spend management interface, that the account number is a valid account number utilizing the list of valid routing numbers.

5. One or more computer-readable non-transitory storage media embodying software that when executed by a processor of a spend management system including a validation database, causes the processor to perform the steps of:

storing a plurality of country block entries in the validation database, wherein each country block entry of the plurality of country block entries includes a bank code parameter and a wire routing number and each of the bank code parameters comprises a regular expression with an associated search pattern;

generating and displaying a spend management interface comprising a plurality of user interface widgets comprising a country widget and a currency widget;

receiving, in the spend management interface, via the country widget, a country identifier from a supplier computer;

receiving, in the spend management interface, via the currency widget, a currency identifier from the supplier computer;

selecting based on the country identifier, a country block entry from the plurality of country block entries and selecting a bank code parameter and wire routing number from the selected country block entry;

autofilling one or more user interface widgets of the plurality of user interface widgets with the wire routing number and bank code parameters and generating, by the spend management interface, a user interface comprising the autofilled widgets;

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displaying the user interface;
 receiving supplier data via the autofilled widgets;
 validating based on the search pattern associated with the
 regular expression of the selected bank code parameter
 and wire routing number, the supplier data received via
 the autofilled widgets; and
 transmitting the validated supplier data to a plurality of
 spend management systems.

6. The one or more computer-readable non-transitory
 storage media of claim 5, wherein the software when
 executed by the processor performs the steps of:
 identifying a currency type, wherein the currency type is
 associated with the currency identifier; and
 determining each of the bank code parameter and wire
 routing parameter in a corresponding country block
 entry based on the currency identifier.

7. The one or more computer-readable non-transitory
 storage media of claim 5, wherein the software when
 executed by the processor performs the steps of:
 determining that the supplier data received via the auto-
 filled widgets is invalid; and

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electronically presenting an error notification to the sup-
 plier computer indicating that the supplier data is
 invalid.

8. The one or more computer-readable non-transitory
 storage media of claim 5, wherein the software when
 executed by the processor performs the steps of:
 displaying one or more hint icons and receiving a hint
 request from the supplier computer, via the displayed
 one or more icons; and
 electronically presenting a suggestion associated with the
 one or more hint icons.

9. The one or more computer-readable non-transitory
 storage media of claim 5, each country block entry of the
 plurality of country block entries further including an
 account number; and wherein the software when executed
 by the processor performs the steps of: storing a list of valid
 routing numbers in the validation database and verifying that
 the account number is a valid account number utilizing the
 list of valid routing numbers.

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